BEHAVIOURAL AUDIOGRAM FROM THE BAT, *MEGADERMA LYRA* (GEOFFROY, 1810; MICROCHIROPTERA)

by

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Abstract

Audition may serve microchiroptera for tasks of active echolocation, communication and the exploitation of acoustical cues from the environment. The relative importance those have for a certain species should also be represented in their audiograms. The audiogram of the Indian false vampire bat (*Megaderma lyra*, **Geoffroy**, 1810) was determined using one variation of a two choice behavioural test in the ultrasonic (20 to 130 kHz) and an other in the sonic range (1 to 22 kHz). Thresholds in the sonic range decrease gradually from a value of 68 dB SPL at 1 kHz to a plateau of about – 5 dB SPL at 16 to 22 kHz with a steep flank of 27 dB/kHz between 15 and 16 kHz. The ultrasonic graph indicates low overall thresholds (– 17 to – 3 dB SPL) leading into a steep flank for frequencies above 90 kHz and with a minor peak (+2.4 dB SPL) at 50 kHz. With the exception of extending to higher frequencies the broadly tuned audiogram resembles those of other small mammals; this is in agreement with the unspecialized cochlea of this species (**Fiedler**, 1983). The above features imply that *M.lyra* is able to make use of its low intensity broadband echolocation call and passive acoustic orientation, respectively. The behavioural significance of the audiogram curve may lie in the peculiar hunting behaviour pattern of *M.lyra* which waits for its prey against dense background, that makes echo clutter exclusion a tremendously difficult task.

Introduction

The audiogram is a common measurement of auditory sensitivity. Microchiropteran bats use the auditory system to cope with several tasks.

In particular, it is an integral part of their sonar system. Therefore, a comparison of threshold intensities in the frequency range of the sonar pulses to the species specific sound characteristics can help to clarify the possible importance of echolocation in a certain species. One part of our study is devoted to that problem.

Moreover, it is of vital interest to pick up acoustic information for communication and for exploitation of other acoustical cues from the environment. In this context, the frequeny range of interest encompasses ultrasonic as well as sonic frequencies. As distant stimuli may be important for the bat, and knowing that atmospheric attenuation is much higher for ultrasonic signals (**Lawrence, Simmons**, 1982), it is important to extend the study to the low frequency range with respect to passive acoustic orientation.

A behavioural approach was chosen for it comes closer to the actual responses to acoustical stimuli in natural surroundings than neurophysiology does. The behavioural audiogram of *M.lyra* was determined for frequencies between 1 kHz and 130 kHz.

Threshold Determination

Experiments in the sonic and ultrasonic range were performed using two different versions of a two choice approach. The data from the two measurement paradigms will be dealt with independently.

Sonic range (1 to 22 kHz, **S.Schmidt**):

5 individuals were trained to orient towards two feeding places in response to pure tone pulses from two loudspeakers.

Viewed from the starting place the two loudspeakers are separated by an angle of 36°; flight distance was 1.55 m. Experiments were performed in a soundproof chamber (Flight cage: 2.6 m x 1.6 m x 1.9 m). Correct responses were rewarded with mealworms. Stimuli were 100 ms pure tone bursts with rise/fall times of 2 ms for $f > 7$ kHz and 10 ms for $f \leq 7$ kHz. The electronic equipment used guaranteed linearity of attenuation and uniform background noise from both loudspeakers.

For threshold determination (see Fig.1) signals were attenuated in 5 dB steps and the percentage of correct responses for each step was plotted against signal intensities. These data were fitted by the optimal Gaussian error function between 50% and 100% correct responses and threshold attenuation was defined as the value corresponding to 75% correct responses. (No. of trials per frequency and animal tested: 123 – 236)

Ultrasonic range (20 to 130 kHz, **B.Türke, B.Vogler**):

The experimental arrangement closely resembles the above described one (Dimensions of the room used: 2.2 m x 5.0 m x 2.28 m; distance between ultrasonic speakers and starting place: 1.88 m; angle between speakers 20°). Stimuli were continous pure tones. Signals in the previously determined critical intensity range were attenuated in 1 – 5 dB steps. When incorrect responses occurred repeatedly at the same signal intensity during one experimental session, this value was taken as a measure of the threshold. Mean thresholds from several (1 to 7) days were determined with the No. of trials per session (8 – 37) taken into account; maximum differences from mean thresholds were 8 dB.

Results

M.lyra has been proven sensitive to frequencies from 1 to 130 kHz (Fig.2).

Thresholds in the sonic range decrease gradually from a value of 68 dB SPL at 1 kHz to a plateau of about – 5 dB SPL at 16 to 22 kHz. The latter is preceded by a less significant fall in sensitivity between 13 and 15 kHz (Data pooled from 5 animals for illustrative purposes; the abovementioned effects can be seen in all individuals to a varying extent).

The ultrasonic graph indicates exceedingly low overall thresholds (–17 to –3 dB SPL) leading into a steep flank for frequencies above 90 kHz; there is a minor peak (+2.4 dB

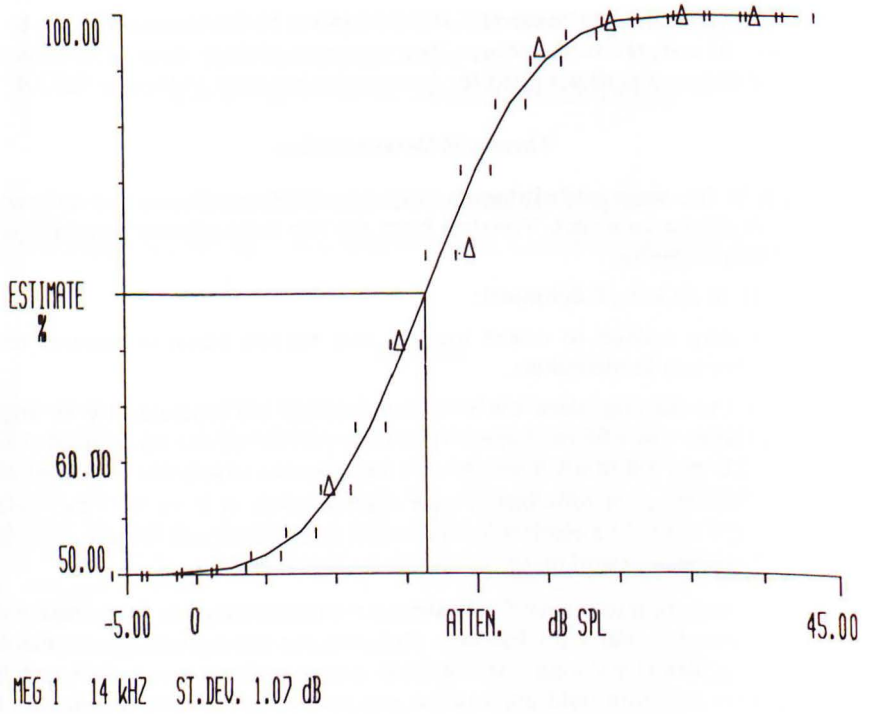


Fig. 1: Threshold determination in bat MEG 1 for F = 14 kHz: The percentage of correct responses (ESTIMATE %) is plotted (upright triangles) versus signal intensities (ATTEN. dB SPL; Scale: 1 unit = 5 dB). The fit gives the optimal Gaussian error function. Dots mark the standard deviation (± 1,07 dB). The horizontal line indicates 75% correct responses, the vertical line gives the corresponding threshold value.

SPL) at 50 kHz (Data presented are from one animal; experiments with a second animal and a series with a modified stimulus give similar values).

Discussion

The behavioural significance of the audiogram curve may be understood by looking at the hunting behaviour of the species. *M.lyra* feeds on frogs, small mammals and insects living on or near the ground (**Habersetzer**, 1983). The dense background causes echo clutter that makes detection of prey by echolocation a very difficult task.

Two solutions are at hand:

According to **Simmons, Fenton, O'Farrell** (1979) a broadband echolocation call maximizes the possible information content of the returning echo.

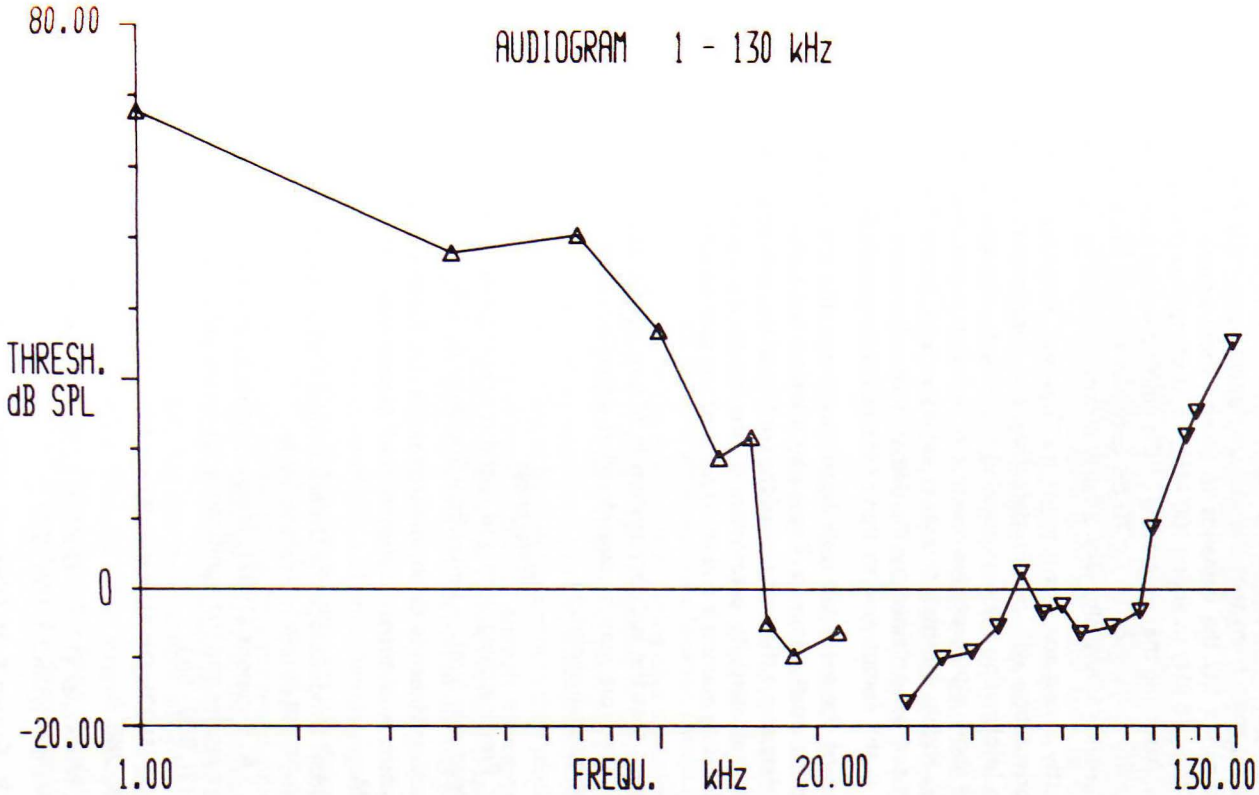


Fig. 2: Threshold intensities (ATTEN. dB SPL) are plotted versus the frequencies (FREQU. kHz) tested. Frequency is drawn in a logarithmic scale. Upright triangles symbolize values obtained for the sonic range. Upside down triangles refer to the ultrasonic range. Threshold values for intermediary frequencies are linearly interpolated. The horizontal bar indicates 0 dB SPL. No. of trials is small for f $>$ 100 kHz; data obtained for 20 kHz with the ultrasonic arrangement (not plotted, No. of trials small) indicate sufficient overlap with the sonic graph.

The sonar pulses of *M.lyra* are broadband signals of low intensity (about 80 dB SPL, **Möhres, Neuweiler,** 1966, **Türke**, unpublished) with a fairly harmonic composition to a first harmonic varying interindividually between 19 kHz and 26 kHz. Pulse duration usually is less than 1 ms. Pulse trains recorded from individuals at rest reveal high variability in the intensity of corresponding harmonic components. The third and fourth harmonic often are most prominent. During flight, there is a slight downward frequency sweep within a pulse; apart from that the spectra remain essentially unchanged. The broadly tuned audiogram fits the demands to receive these signals: It indicates high hearing sensitivity from 16 kHz to about 100 kHz, i.e. it comprises the frequency range of the sonar pulses. Assuming that processing of the emitted sound and its echo do not interfere, the intensity difference of $>$70 dB between emitted pulses and auditory thresholds should provide *M.lyra* also with a fairly sensitive means for echolocation.

M.lyra can efficiently orient towards a target by "passive" listening (**Fiedler**, 1979). Passive acoustic orientation will permit detection and localization of hidden prey and allows to gain information without warning the prey by the echolocation signal. With regard to the above signal, passive acoustic orientation seems to be an especially promising strategy: Highly sensitive threshold values extend farther to the low frequency range than can be explained by the demands of echolocation. Below 16 kHz hearing sensitivity is still comparable to that of the housemouse (**Markl, Ehret,** 1973).

With the exception of the extended high frequency range, the audiogram of *M.lyra* resembles that of other small mammals. These physiological and behavioural results are in agreement with data about the anatomy of the cochlea of this species (**Fiedler**, 1983). The relative length and elasticity parameters of the basilar membrane are similar to those in non-echolocating mammals and show none of the specializations found in bats with CF–FM calls.

These findings suggest that the auditory system in *M.lyra* has features of non-echolocating mammals. It therefore may be considered as a starting point for further specialization in microchiropteran evolution.

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